

CS5483: DATA WAREHOUSING AND DATA MINING

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Data Warehousing and Data Mining

Subject Code

CS - Computer Science

Course Number

5483

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS3402 Database Systems or
CS5481 Data Engineering or equivalent

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

The course introduces the subject of automatic/semi-automatic knowledge discovery from data. It gives an overview of the principles and applications of supervised/unsupervised learning, and reinforces the concepts with hands-on experience using some state-of-the-art software. The students will learn to apply different learning algorithms and evaluate them with appropriate statistical performance metrics and techniques that avoid overfitting. Students will also learn how corporations manage their data with dimensional models and OLAP for simple and fast data analysis.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Identify and explain the main characteristics of different data warehousing and data mining techniques through observation of their operations.		x		
2	Critically evaluate the strengths and limitations of current data warehousing and data mining techniques.		x		
3	Apply the main algorithms in data warehousing and data mining in a computationally efficient way.			x	
4	Design new solutions for data warehousing and data mining problems by improving and combining current techniques.			x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will learn data warehousing and data mining techniques, and their applications in different problem domains.	1, 2	2 hours/ week

2	Tutorial	Students will work on a set of problems on the principles and applications of data warehousing and data mining, and present their solutions in the class.	1, 2	1 hour/ week
3	Project	Students will complete two projects. The first project gives students an opportunity to step through the process of knowledge discovery from data. The second project requires students to gain deeper insights of the existing solutions and encourage them to create new designs.	3, 4	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Project 1 (Implementation of data mining algorithms.)	3	15	-	Yes
2	Project 2 (Application of data warehousing and data mining techniques to real world problems.)	4	15	-	Yes
3	Quiz	1, 2	20	-	No

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

1.1 Capacity for effectively implementing data warehousing and data mining algorithms in a computationally efficient way.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

1.2 Capability to create new solutions for data warehousing and data mining problems by improving and combining different techniques.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

2.1 Ability to explain in detail the principles of different data warehousing and data mining techniques

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

2.2 Capability to correctly apply suitable techniques to solve data warehousing and data mining problems.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

3.1 Capacity for explaining the main characteristics of different data warehousing and data mining techniques in depth.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

1.1 Capacity for effectively implementing data warehousing and data mining algorithms in a computationally efficient way.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

1.2 Capability to create new solutions for data warehousing and data mining problems by improving and combining different techniques.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

2.1 Ability to explain in detail the principles of different data warehousing and data mining techniques

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

2.2 Capability to correctly apply suitable techniques to solve data warehousing and data mining problems.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

3.1 Capacity for explaining the main characteristics of different data warehousing and data mining techniques in depth.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

Data Mining:

Supervised learning: classification, numerical prediction, decision-trees.

Unsupervised learning: association rule mining clustering.

Performance evaluation: stratification, cross-validations, overfitting.

Data Warehousing:

Dimension modeling, star schema, on-line analytical processing (OLAP)

Reading List

Compulsory Readings

Title	
1	Han J. and Kamber M. Data Mining: Concepts and Techniques 3rd Ed., Morgan Kaufmann (2011).
2	Witten I., Frank E. and Hall M. Data Mining: Practical Machine Learning Tools and Techniques, 3rd Ed., Morgan Kaufmann (2011).

Additional Readings

Title	
1	Hastie T., Tibshirani R. and Friedman J. The Elements of Statistical Learning, 2nd Ed., Springer (2009).
2	VanderPlas, Jake. Python data science handbook: Essential tools for working with data. "O'Reilly Media, Inc.", 2016.
3	Kimball R. and Ross M. The Data Warehouse Toolkit, 3rd Ed., Wiley (2013).